



Live Trading Using IB API

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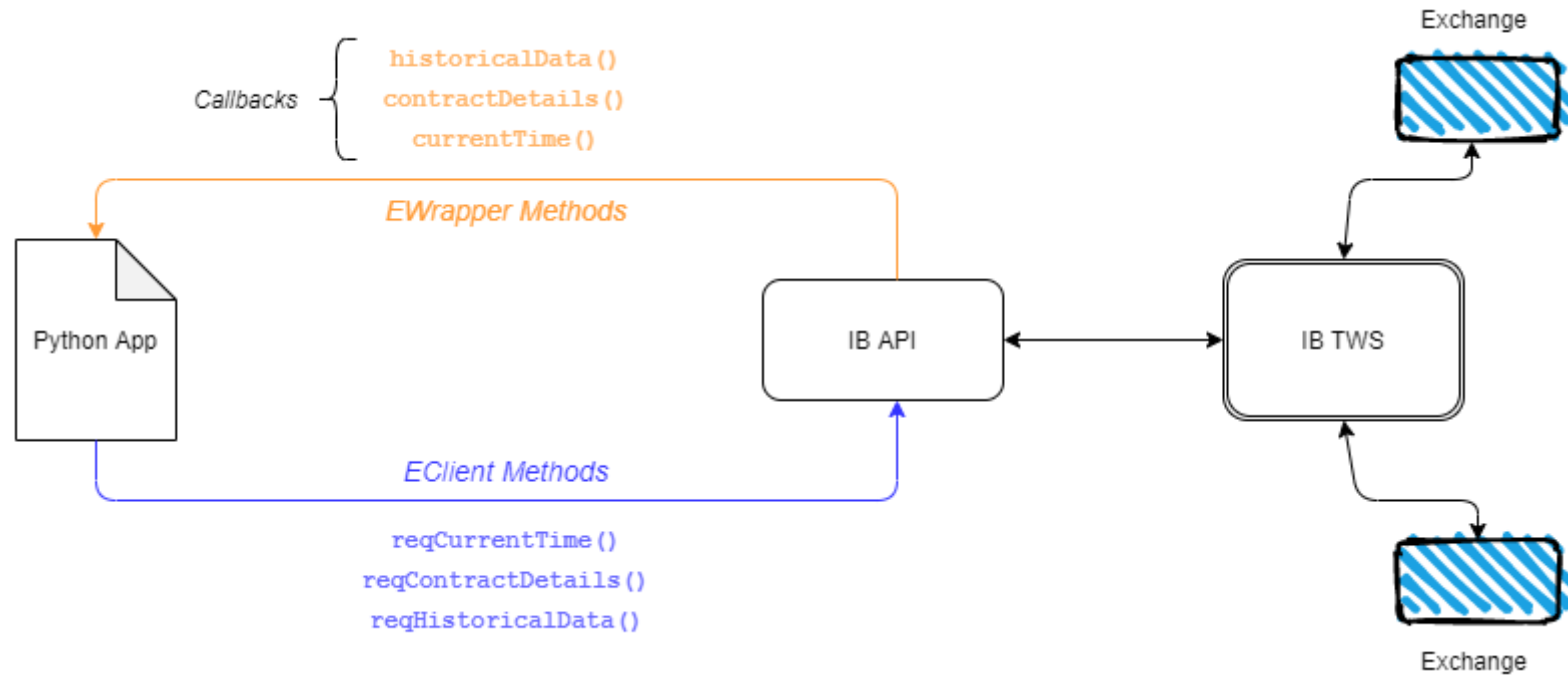
What are we are trying to achieve?

- Develop a full-fledged intraday strategy using IB API

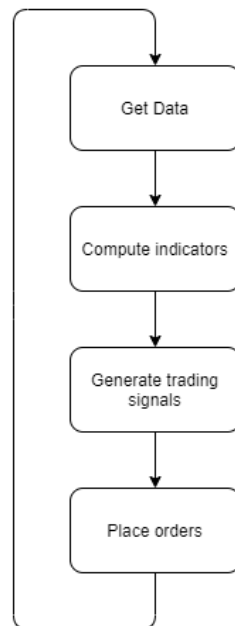
How will we achieve the objective?

- Define strategy components
- Dissect each component
- Understand the code structure
- Demo of the strategy
- Limitations & enhancements

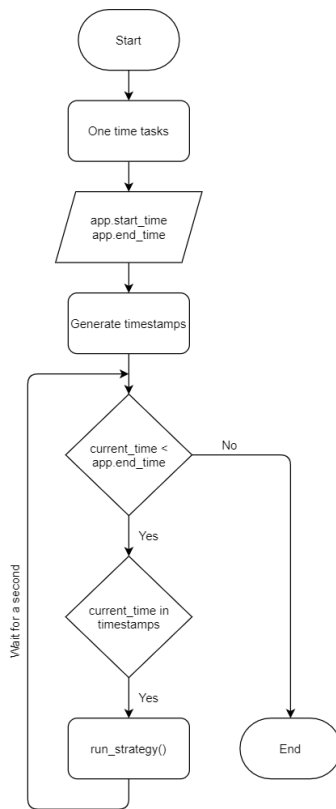
Recap - EClient & EWrapper Classes



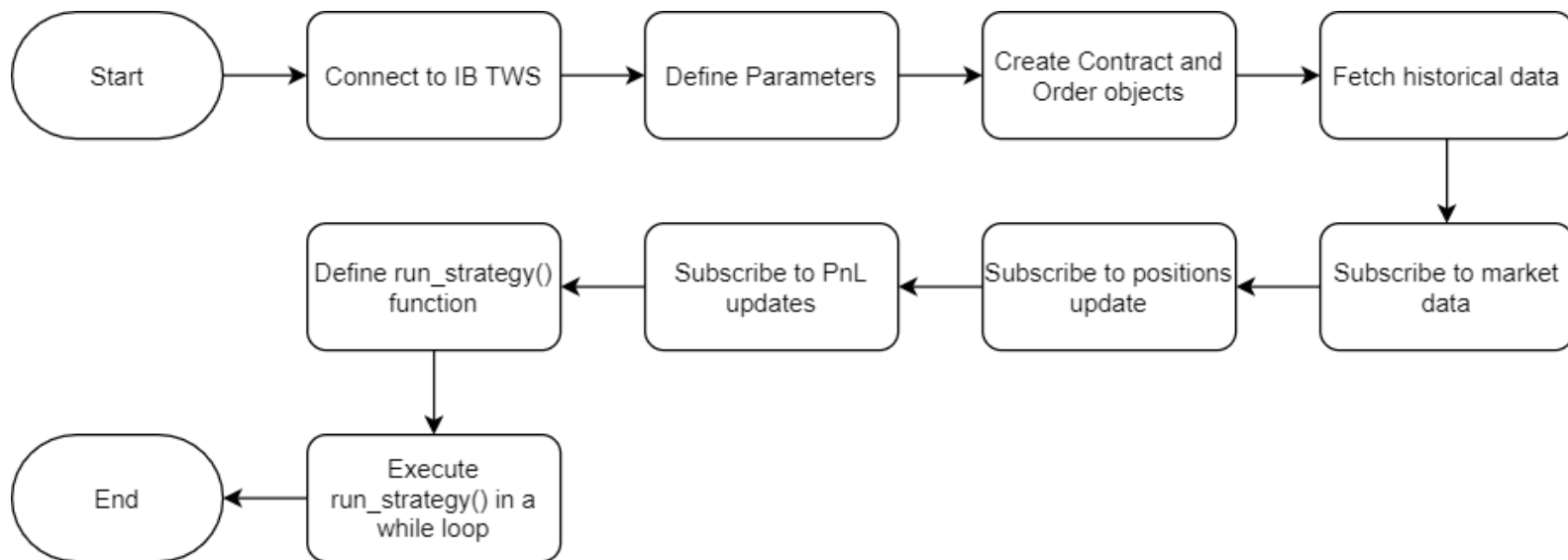
The core of the strategy - `run_strategy ()`



Scheduling part - while loop



All components



Defining parameters

- Strategy parameters
 - Lookback period for mean and standard deviation
 - Number of standard deviations for upper and lower bands
- Strategy start and end time
- Trading frequency
- Resampling frequency
- Assets to trade
- Order parameters
 - Order type
 - Quantity

One time tasks

- Define objects for the trading instrument and order
- Fetch historical data
 - We will fetch it only once when the strategy starts
- Subscribe to market data
- Subscribe to position updates
- Subscribe to PnL updates
- Define `run_strategy()` method
- Implement while loop to execute strategy

Resample data

- Fetch historical data for once
- Resample tick data on every iteration and append it to the historical data

Compute indicators

- Fetch the last n data points from the data
- Compute the required variables
 - Rolling mean
 - Rolling standard deviation
 - Upper band
 - Lower band

Generate trading signals

- Enter long if the current price is below the lower band $\rightarrow 1$
- Enter short if the current price is above the upper band $\rightarrow -1$
- Exit long if the current price crosses the mean value from below $\rightarrow 0$
- Exit short if the current price crosses the mean value from above $\rightarrow 0$

Generate trading signals

- Place market orders based on the trading signals generated
- If a buy signal is generated:
 - If we don't have any existing long position:
 - Go long
 - If we already have a short position:
 - Exit short position
- If a sell signal is generated:
 - If we don't have any existing short position:
 - Go short
 - If we already have a long position
 - Exit long position

*It's **Demo** time!*

Things to note

- No reconciliation logic
- Fires market orders
- Logging not available
- Single instrument being traded

Thank you!